

Patient Shortage in Histopathology: Does Unequal Patient Influence Explain the Damage?

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Abstract

Reducing the number of training patients can impair classification even when class patch counts remain fixed. This experiment tests whether equalizing patient influence reduces that damage. Multinomial logistic regression is compared under patch-average and patient-average training on identical concentrated and spread allocations from BRACS and TCGA-UT. Frozen features, patient-disjoint splits, and regularization remain fixed. Under concentrated support, patient averaging changes patient-macro balanced accuracy by -1.36 percentage points on BRACS (95% interval: -3.09 to 0.18) and -0.004 points on TCGA-UT (-0.017 to 0.009). Broader coverage still improves patient-average training by 2.73 and 4.70 points, respectively. The contribution audit distinguishes these outcomes: weighting substantially redistributes loss mass on BRACS, whereas TCGA-UT already has nearly equal patient contributions. Equalizing influence therefore leaves the established coverage deficit intact in these allocations. The remaining benefit of additional patients motivates examining which variation broader coverage supplies.

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1 Motivation

The signal taxonomy of Queisler (2026c) distinguishes nominal support, the number of labelled examples, from independent support, the number of patients supplying them. Patches from one patient share biological and acquisition characteristics, so their count need not reflect the amount of independent evidence. Within-class diversity describes the variation observed under a specified representation. Together, these concepts explain why a large patch collection can still provide limited exposure to differences between patients.

The controlled shortage benchmark of Queisler (2026b) reduced patient support while holding class patch counts fixed. Discrimination losses reached 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT, and tested methods left these losses largely unrecovered. The classifier comparison of Queisler (2026d) evaluated the original multilayer perceptron alongside logistic regression and cosine nearest neighbours. Under concentrated support, logistic regression improved patient-macro balanced accuracy by only 0.19 points on BRACS and 0.48 points on TCGA-UT. Broader coverage benefited logistic regression by 2.44 points (95% interval: 0.31 to 4.98) and 4.70 points (4.04 to 5.34), respectively. These findings support a coverage limitation within the tested procedures and motivate examining how patient shortage affects learning.

Unequal patient influence arises because ordinary training assigns equal weight to patches. Within a class, a patient supplying more patches consequently contributes more of the training loss. When few patients supply the available patches, the fitted classifier may depend strongly on the characteristics of those with the largest contributions. Those characteristics may be less representative of patients encountered at evaluation. Patient-average training redistributes the loss so that each contributing patient has equal weight within a class. A gain on the same training patches would support unequal influence as a contributor to the observed damage. Here, influence refers to loss weight; the observations themselves still determine how each patient affects the fitted decision rule.

Broader patient coverage also exposes the classifier to additional variation within each class, including differences in tissue appearance and acquisition conditions. Even when all observed patients receive equal weight, a small patient set may represent this variation incompletely. A *residual coverage limitation* is the performance benefit of broader coverage that remains after patient averaging. This residual describes how much of the coverage deficit survives the correction. Unequal influence and incomplete coverage can therefore contribute together: weighting may recover part of the loss while additional patients remain beneficial.

The central question is whether equalizing patient influence reduces the established coverage deficit. The comparison measures both the gain under concentrated support and the remaining benefit of spread support. The benchmark’s round-robin sampling and contribution caps already limit unequal influence (Queisler, 2026a). The observed contribution distribution therefore establishes how strongly patient averaging changes the training objective in these allocations.

2 Study design

BRACS retains its seven lesion classes and TCGA-UT its 30 eligible cancer types. Both retain the benchmark’s eligibility rules, class order, and frozen 2560-dimensional Virchow2 features, formed by concatenating the CLS token and mean patch token. The three existing train–validation–test splits remain fixed. BRACS patients and TCGA-UT participants are called patients below; they are disjoint between partitions within each split (Queisler, 2026a).

Concentrated support (C) and spread support (S) use the exact balanced training allocations evaluated by Queisler (2026d): the shared concentrated allocation and the spread allocation under native class assignment. Class-specific patch counts match between support conditions

within each split. The two objectives receive identical patches within each condition. Validation and test cohorts retain their natural distributions.

Patient support	Patch-average loss	Patient-average loss
Concentrated	Existing logistic baseline	Weighted fit on identical patches
Spread	Existing logistic baseline	Weighted fit on identical patches

Table 1: Four cells per dataset and split compare two objectives under two support conditions, isolating the weighting intervention within each condition.

The inherited allocations contain 13,982–27,202 patches per BRACS split, with 78–89 concentrated patients and 103 spread patients. TCGA-UT contains 36,988–52,204 patches, with 671–917 concentrated patients and 4,983 spread patients (Queisler, 2026d). These unequal shortage doses prevent attributing differences in effect size to intrinsic dataset susceptibility. The support contrast also changes patient identities and within-patient patch contributions; it does not isolate patient count alone.

3 Patient-average training

For one training allocation, N denotes its total patch count, n_c the count for class c , and G_c the number of patients contributing to that class. Patient i contributes $m_{ic} > 0$ such patches, whose indices form $\mathcal{J}_{ic}$. Thus $n_c = \sum_{i=1}^{G_c} m_{ic}$. Logistic regression predicts class probabilities from $Wz_j + b$, where z_j is a frozen feature vector, W the coefficient matrix, and b an unpenalized intercept. Its cross-entropy for patch j is $\ell_j = -\log p(y_j | z_j)$.

Ordinary training gives each patch equal weight. Patient-average training instead uses

$$w_j = \frac{n_c}{G_c m_{ic}} \quad (j \in \mathcal{J}_{ic}), \quad L_{\text{patient}} = \frac{1}{N} \sum_{j=1}^N w_j \ell_j + \frac{\lambda}{2} \|W\|_F^2. \quad (1)$$

Each patient then contributes total weight n_c/G_c within class c . Equivalently, the data term averages patch losses within a patient, averages patients within a class, and combines classes with their unchanged shares n_c/N :

$$\frac{1}{N} \sum_j w_j \ell_j = \sum_{c=1}^K \frac{n_c}{N} \frac{1}{G_c} \sum_{i=1}^{G_c} \frac{1}{m_{ic}} \sum_{j \in \mathcal{J}_{ic}} \ell_j. \quad (2)$$

Here K is the number of classes. Preserving n_c/N also preserves small integer-rounding differences in the balanced allocations. A patient contributing several classes enters each class separately; the objective does not force equal total influence across such patients irrespective of class.

The weights satisfy $\sum_{j: y_j=c} w_j = n_c$ and $\sum_j w_j = N$. Their mean is one, so weighting does not change the loss scale relative to regularization. If all patients contribute equal numbers of patches within each class, every weight equals one and the two objectives coincide.

3.1 Fixed fitting procedure

Both objectives use the deterministic multinomial logistic regression procedure of Queisler (2026d), with the same feature preprocessing and numerical convergence criteria. Regularization remains fixed at the validation-selected values in Table 2. The analysis includes only converged fits, so the comparison concerns the weighting intervention under matched fitting conditions.

Dataset	Concentrated λ	Spread λ
BRACS	1	0.1
TCGA-UT	0.01	0.01

Table 2: Validation-selected regularization from Queisler (2026d), held fixed for both objectives across all three splits.

Prediction selects the largest logit, resolving ties by the fixed class order. Validation and test patients never enter fitting or weight calculation. Freezing regularization isolates the weighting change within each support condition. On BRACS, the coverage contrast remains a comparison of procedures with different inherited regularization; it is not a fixed-regularization patient-count effect.

3.2 Patient contributions

The contribution distribution is characterized before fitting by n_c , G_c , the minimum and maximum m_{ic} , and the largest patient share $\max_i(m_{ic}/n_c)$, for every class, split, and support condition. Departure from equal influence is measured by

$$D_c = \frac{1}{2} \sum_{i=1}^{G_c} \left| \frac{m_{ic}}{n_c} - \frac{1}{G_c} \right|. \quad (3)$$

This is the fraction of within-class loss mass redistributed by patient averaging; zero means the objectives already coincide for that class. The measure describes inequality in patients’ contributions to the training objective. Allocation and weight checks establish identical patches across objectives, matching class patch counts across support conditions, positive weights, preserved class loss mass, total weight N , and unit weights under equal contributions. The benchmark’s 10% patient and 5% slide caps constrain patch allocation; patient-average loss weights follow Equation (1).

4 Evaluation

The primary endpoint is *patient-macro balanced accuracy*, with equal weighting of patients within each class as in Queisler (2026d). For split r , $\mathcal{P}_{rc}$ contains test patients with at least one patch of true class c . Their within-patient class recalls are q_{irc} . The split score is

$$A_r = \frac{1}{K} \sum_{c=1}^K \frac{1}{|\mathcal{P}_{rc}|} \sum_{i \in \mathcal{P}_{rc}} q_{irc}, \quad q_{irc} = \frac{1}{|\mathcal{J}_{irc}^{\text{test}}|} \sum_{j \in \mathcal{J}_{irc}^{\text{test}}} \mathbf{1}\{\hat{y}_j = c\}. \quad (4)$$

Here $\mathcal{J}_{irc}^{\text{test}}$ contains that patient’s evaluated patches of class c . The final score averages the three split scores equally, separately for each dataset. Secondary endpoints are class-specific patient-macro recalls and patch-micro balanced accuracy, which pools patches within class before averaging classes. Probability quality and calibration are outside this experiment’s scope.

The pooled score for objective $o \in \{\text{patch}, \text{patient}\}$ under support $s \in \{\text{C}, \text{S}\}$ is denoted by $A_{o,s}$. The contrasts are

$$W_s = A_{\text{patient},s} - A_{\text{patch},s}, \quad \text{weighting gain}, \quad (5)$$

$$B_o = A_{o,S} - A_{o,C}, \quad \text{coverage benefit}, \quad (6)$$

$$I = W_C - W_S = B_{\text{patch}} - B_{\text{patient}}, \quad \text{interaction}. \quad (7)$$

The primary questions concern W_C and B_{patient} . All four accuracies are reported as percentages; both weighting gains, both coverage benefits, and the interaction are expressed in percentage points. A positive interaction means the coverage gap shrinks, but could arise because spread performance deteriorates. Its interpretation therefore depends on the absolute scores and both gains.

Uncertainty is estimated with 2,000 paired Bayesian-bootstrap replicates over distinct test patients, following Queisler (2026d). Each replicate assigns random positive patient weights from a $\text{Dirichlet}(1, \dots, 1)$ distribution. Each patient’s weight is shared across objectives, support conditions, classes, and repeated test appearances across splits. Within-class recalls are recalculated as weighted patient means before averaging splits and forming contrasts. The 2.5th and 97.5th percentiles define individual exploratory 95% intervals, accompanied by split-level estimates. The intervals describe patient variation conditional on the deterministic fits; they are not adjusted for multiple comparisons.

4.1 Interpretation criteria

One percentage point is the prespecified practical-relevance threshold, following Queisler (2026d). A point estimate above one point is practically promising; a lower interval bound above one point supports a positive effect exceeding that threshold. Intervals crossing zero remain inconclusive about direction. An interval excluding zero but overlapping one point supports direction without establishing a practically meaningful gain. A negligible residual coverage difference requires the entire interval for B_{patient} to lie within $[-1, 1]$ points.

Weighting gains are interpreted together with the patient contribution distribution. Near-zero D_c means the intervention changes little in that class, so a null result provides little evidence about strongly unequal patient influence elsewhere. Similar gains in both support conditions indicate a general weighting benefit; larger gains under concentrated support suggest a contribution specific to patient shortage. Broad or conflicting intervals leave the mechanism unresolved.

5 Results

Patient averaging does not reduce the established coverage deficit. Its pooled accuracy changes are negative on BRACS and close to zero on TCGA-UT, while spread support remains beneficial. The contribution distributions show that these outcomes arise under substantially different degrees of initial patient inequality.

5.1 Realized patient contributions

All twelve training-allocation audits passed, including matching class patch counts across support conditions, baseline compatibility, and preservation of class and total loss mass. All twelve patient-average fits converged. The regularization values in Table 2 were retained. Each BRACS split contains 24 validation and 24 test patients; each TCGA-UT split contains 1,067 in each partition. These are per-split counts, with possible patient overlap across repeated splits.

Table 3 summarizes the realized contribution distributions. BRACS exhibits appreciable inequality despite the allocation caps. Under concentrated support, D_c ranges from 4.27% to 36.31%, with an unweighted mean of 20.41% across class-split combinations. Under spread support, the range is 2.74%–35.58% and the mean is 17.42%. Patient averaging therefore produces a substantial intervention in both conditions.

TCGA-UT contributions are much closer to equal. Concentrated-support D_c ranges from zero to 3.85%, with a mean of 0.48%. Spread-support D_c ranges from zero to 9.99%, with a mean of 1.99%. A relatively large maximum patient share under concentrated support can therefore reflect few contributing patients rather than substantial inequality among them. The full class- and split-specific contribution summaries accompany the report as supplementary data.

Dataset	Support	Patients G_c	Patches m_{ic}	Max. share (%)	D_c (%)
BRACS	C	20–49	1–375	9.65	4.27–36.31
	S	30–69	1–339	8.72	2.74–35.58
TCGA-UT	C	20–44	20–154	8.85	0.00–3.85
	S	30–514	2–58	3.41	0.00–9.99

Table 3: Contribution ranges across classes and splits. Patch counts refer to individual patients within a class. Maximum share is the largest observed fraction of a class’s patches supplied by one patient. D_c , from Equation (3), measures the fraction of class loss mass redistributed by patient averaging.

5.2 Weighting gains and residual coverage benefits

On BRACS, patient averaging reduces concentrated-support accuracy from 44.99% to 43.63% and spread-support accuracy from 47.43% to 46.36% (Table 4). The corresponding changes are -1.36 and -1.08 points. Both intervals include zero, so neither establishes deterioration. However, their upper bounds, 0.18 and 0.42 points, are below the one-point threshold for a practically meaningful improvement.

Dataset	Objective	Accuracy C (%)	Accuracy S (%)	Benefit (points)
BRACS	Patch-average	44.99	47.43	2.44
		[41.89, 48.13]	[43.55, 51.18]	[0.31, 4.98]
	Patient-average	43.63	46.36	2.73
		[40.57, 46.62]	[43.15, 49.58]	[0.67, 5.15]
TCGA-UT	Patch-average	75.15	79.84	4.70
		[73.77, 76.39]	[78.50, 81.08]	[4.04, 5.34]
	Patient-average	75.14	79.84	4.70
		[73.76, 76.38]	[78.49, 81.08]	[4.04, 5.32]

Table 4: Patient-macro balanced accuracy and coverage benefit $B_o = A_{o,S} - A_{o,C}$, averaged equally across three splits. Brackets contain paired-bootstrap 95% intervals. Benefits are calculated before rounding the component accuracies.

Broader coverage still benefits patient-average training by 2.73 points on BRACS (95% interval: 0.67 to 5.15). The interval supports a positive benefit but does not establish that it exceeds one point. The interaction is -0.29 points (-1.88 to 1.33), providing no supported shortage-specific gain from weighting. The point estimate of the coverage gap is slightly larger after patient averaging.

Dataset	Weighting gain W_C	Weighting gain W_S	Interaction I
BRACS	-1.363	-1.077	-0.286
	$[-3.089, 0.175]$	$[-3.183, 0.421]$	$[-1.884, 1.325]$
TCGA-UT	-0.004	-0.007	0.004
	$[-0.017, 0.009]$	$[-0.030, 0.013]$	$[-0.021, 0.031]$

Table 5: Paired weighting effects in percentage points, with individual 95% intervals. Three decimal places retain the small TCGA-UT effects. Positive values favour patient averaging; a positive interaction indicates a larger gain under concentrated support.

On TCGA-UT, both objectives achieve almost identical accuracy. Concentrated-support weighting changes accuracy by -0.004 points (95% interval: -0.017 to 0.009), and spread-support weighting by -0.007 points (-0.030 to 0.013). Both intervals lie entirely within ± 1 point. The residual coverage benefit is 4.70 points (4.04 to 5.32), virtually unchanged from the patch-average

benefit. The near-zero interaction likewise provides no indication that weighting closes the coverage gap. Neither dataset meets the criterion for a practically negligible residual coverage difference.

5.3 Split consistency and secondary outcomes

Coverage benefits under patient averaging are positive in all three splits of both datasets (Table 6). BRACS concentrated-support weighting gains are negative in every split, while spread-support gains change sign. TCGA-UT weighting changes remain within 0.05 points of zero in each split and condition. The much wider range of BRACS coverage effects reinforces the uncertainty in its pooled estimate.

Dataset	Split	W_C	W_S	B_{patch}	B_{patient}
BRACS	0	-1.28	-2.14	2.45	1.59
	1	-0.10	0.18	0.27	0.55
	2	-2.70	-1.27	4.60	6.04
TCGA-UT	0	0.008	0.014	4.31	4.32
	1	-0.008	0.005	4.68	4.69
	2	-0.011	-0.041	5.10	5.07

Table 6: Split-level point estimates in percentage points. The split identifiers refer to the fixed patient-disjoint partitions. The splits can share patients and do not constitute independent cohort replications.

Patch-micro balanced accuracy shows the same absence of recovery. On BRACS, concentrated-support accuracy changes from 50.21% to 49.38%, and spread-support accuracy from 50.04% to 49.73%. On TCGA-UT, accuracy rounds to 75.77% under concentrated support and 80.07% under spread support for both objectives; the underlying changes are -0.001 and -0.007 points.

BRACS class-level weighting gains are mixed. Under concentrated support, flat epithelial atypia and atypical ductal hyperplasia gain 1.92 and 1.93 points, while usual ductal hyperplasia loses 4.95 points. Under spread support, the largest class decline is 5.12 points for ductal carcinoma in situ. On TCGA-UT, all split-averaged class changes lie between -0.225 and 0.085 points under concentrated support, and between -0.122 and 0.083 points under spread support. These are descriptive secondary point estimates without class-specific uncertainty intervals. Complete class recalls for both objectives accompany the report as supplementary data.

6 Discussion

Equalizing patient influence does not recover the observed patient-shortage damage. On BRACS, the correction materially changes the loss distribution but provides no supported accuracy gain. On TCGA-UT, the correction is small because the training allocations already distribute patches nearly equally among patients. Broader coverage remains beneficial under both objectives. Together, these findings narrow the explanation of the established deficit: unequal contribution to the training loss is not sufficient to account for it under the tested procedure.

The distinction between the datasets matters. BRACS supplies the more substantial intervention on unequal influence: concentrated-support weighting redistributes an average of 20.4% of within-class loss mass. Its negative accuracy estimate does not establish harm because the interval includes zero, but the upper bound of 0.18 points excludes a one-point improvement within this conditional analysis. Equal patient weights can also increase the influence of patients represented by few patches. That change may emphasize less representative observations

or less precisely sampled patient characteristics. These are possible explanations of the negative estimate, not mechanisms established by this comparison.

TCGA-UT provides especially clear evidence that a coverage deficit can persist when patient contributions are already almost uniform. The concentrated allocation requires redistribution of only 0.48% of within-class loss mass on average, and both weighting-gain intervals lie within ± 1 point. Yet the residual coverage benefit is 4.70 points, with its lower interval bound above four points. This combination weakens unequal patient influence as an explanation of the existing TCGA-UT deficit. It gives limited evidence about what patient averaging would do in a different allocation with strongly unequal contributions.

The coverage benefit remains less precisely estimated on BRACS. The patient-average interval excludes zero, and the point estimate exceeds the practical threshold, but the interval does not establish a benefit larger than one point. This distinction prevents treating the two datasets as equally strong confirmations. Neither dataset supports a practically negligible residual coverage difference, and neither interaction establishes that weighting preferentially repairs shortage damage.

The persistent coverage benefit motivates testing which observations additional patients supply. A follow-up could compare random and feature-diverse patient selection at matched patient counts, class patch counts, and contribution limits, with repeated patient draws. Comparing added patients with added patches from existing patients could separately examine within-patient redundancy. These comparisons would help distinguish patient composition and sampling depth as explanations of the residual deficit.

7 Limitations

This experiment is conditional on the exact allocations and cohorts that motivated the hypothesis. Its intervals describe held-out patient variation, not variation from new training-patient draws, hyperparameter selection, or another population. Shared bootstrap weights account for repeated test patients but do not remove dependence from overlapping training and validation partitions. Fixed regularization tests this intervention at the inherited settings, not the best attainable patient-average procedure.

A residual coverage benefit cannot distinguish missing morphology, acquisition variation, patient composition, or within-patient sampling. Patient averaging changes influence without removing patch redundancy. Neither a residual deficit nor a null gain establishes an information ceiling, intrinsic inseparability, or classifier impossibility. Feature diversity, patch depth, fresh allocations, additional readouts, and other representations require separate experiments.

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